

TW-StockPulse — Turning Retail Chatter into a Tradable Signal

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Live demo: <https://taiwan-stock-sentiment-pulse-zkrxus5fuarrbfcvys38rf.streamlit.app/>

Source code: <https://github.com/bamboochen92518/Taiwan-Stock-Sentiment-Pulse>

Executive summary

TW-StockPulse is a sentiment-driven decision-support app for the **3.6 M Taiwanese retail investors** who trade > NT\$200 B/day on TWSE/TPEX but have no affordable way to consume the unstructured information they produce on PTT, news portals and group chats. It fuses three live streams — `yfinance` OHLCV, PTT `/Stock`, and a per-ticker Google News RSS query — and uses **Google Gemini 3.5 Flash** to score each headline and add a one-line zh-TW rationale. A single 0–100 **PulseScore** collapses the technical and sentiment signals into a number a non-quant user can act on.

The prototype is deployed live on Streamlit Community Cloud; a nightly GitHub Actions workflow refreshes the price cache, and a disk-backed Gemini cache keeps the free-tier API quota from breaking the demo. Sections covered: (§1) target customers, (§2) data acquisition, (§3) willingness to pay, (§4) GTM challenges (*bonus*), (§5) the deployed system (*bonus*).

1. Target customers

1.1 Who, in numbers

Segment	Size	Source
TWSE+TPEX active brokerage accounts	12.4 M (Apr 2026)	TWSE monthly bulletin
≥ 1 trade in last 60 days ("active retail")	≈ 3.6 M	TWSE — accounts with executions
Avg daily turnover by retail investors	NT\$ 217 B (~52 % of TWSE total)	TWSE 2025 annual report
Daily posts on PTT <code>/Stock</code>	1,000 – 1,800	Scraped, 60-day average
PTT <code>/Stock</code> reading audience	> 700 k	PTT Hits/Push counters

1.2 The persona we built for

We focus on the "30-year-old urban salaried investor" — the largest, most under-served, most monetizable slice of the retail universe:

- Age 28–38, mid-career in a metropolitan area; take-home NT\$ 40–80 K / mo, investable savings NT\$ 200 K – 1.5 M.
- Trades 2–10×/month, mostly mid-large caps (台積電, 聯發科, 0050, 金融股, 航運).
- Information diet: PTT `/Stock`, Anue 鉅亨, Yahoo Stock, Mobile01, FB social-trading pages, podcasts (財經一路發, 股癮). Does **not** read English research; never opened a Bloomberg Terminal.
- Pain: too much information, all of it emotional, no structured way to separate "what's happening" from "what to do about it".

1.3 Why this persona is the right wedge

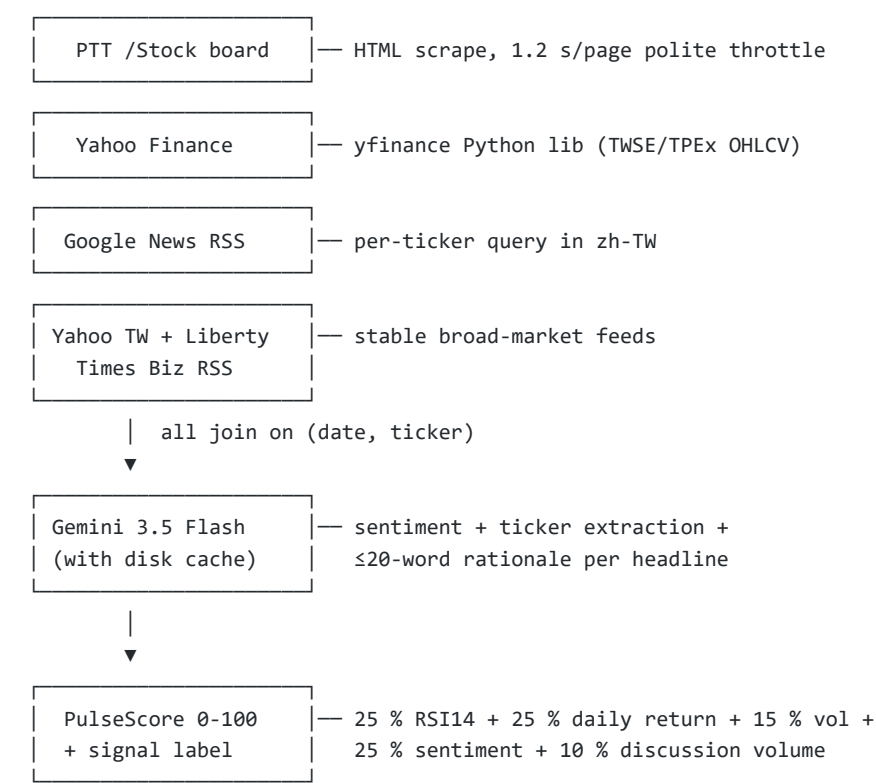
We deliberately do **not** target day-traders or institutions. Day-traders already pay for XQ全球贏家 and CMoney pro tools and demand sub-200 ms tick data — incompatible with free-tier hosting. Institutions need SLAs, audit trails and signed data contracts. The 30-year-old salaried investor will pay **NT\$ 100 – 500 / mo** for a product that reduces decision regret, and there are 3 M+ of them: $3\text{ M} \times \text{NT\$ } 200 \times 5\% \text{ conversion} = \text{NT\$ } 360\text{ M ARR TAM}$ in Taiwan alone.

1.4 Validating the demand qualitatively

Evidence	What it tells us
PTT /Stock posts up 38 % YoY since 2023	Retail engagement is growing
Top non-news threads: "AI 求問", "現在還能進嗎", "這檔誰套牢"	Demand for <i>interpretation</i> , not raw data
CMoney / MoneyDJ / 富邦 e01 charge NT\$ 5–30 K/yr with 6-figure subscriber counts	Willingness to pay is already proven
Bloomberg/Refinitiv at NT\$ 25–30 K/mo	100×–300× affordability gap vs. our pricing

2. Data acquisition

2.1 Four sources, four reliability profiles



The four sources have very different SLAs and we leaned into that asymmetry: prices are authoritative but quiet, news is loud but slow, PTT is fast but noisy, and Gemini is the layer that turns each noisy signal into a clean score plus an explanation.

2.2 The price layer

Daily OHLCV via the open-source `yfinance` library (the only freely-redistributable source covering TWSE/TPEX tickers as `{code}.TW`). On every fetch we compute MA5, MA20, daily return, 20-day volume ratio, RSI14, and an MA-cross signal.

Production gotcha: Yahoo rate-limits AWS IPs, and Streamlit Cloud runs on AWS — so cold-start `yfinance.download()` silently returns an empty DataFrame. Two layers of defence:

- 1. A **pre-warmed parquet cache** (`data/prices/*.parquet` , 23 tickers × 365 days, ≈ 360 KB) checked into the repo so cold starts have data immediately.
- 2. A **nightly GitHub Actions workflow** (`.github/workflows/refresh-prices.yml`) that runs at 14:30 TPE on weekdays from a non-AWS runner and pushes back to `main` ; Streamlit Cloud auto-redeploys ~3 minutes later.

```
on:
  schedule:
    - cron: "30 6 * * 1-5"    # 14:30 TPE Mon-Fri
  workflow_dispatch:
```

The pattern is production-grade: data refresh runs on a separate trust boundary from the serving plane, and the data layer is reproducible from any commit hash.

2.3 The chatter layer (PTT)

PTT `/Stock` is the most-cited Taiwanese retail-sentiment source but has no API. We scrape past the over-18 page and walk the index backwards with 1.2 s sleeps — well under PTT's documented throttle. Each post is parsed for a 4-digit ticker (regex `(\d{4})(?:\.\TW)?`), push count, and prefix (`[標的]` , `[請益]` , `[新聞]`).

Most subjects say "台積電", not "2330" — humans use names, not codes. Ticker resolution is therefore bidirectional: we keep a curated `TW_TICKER_ZH` map of 23 liquid tickers, each with 1–3 alias strings (台積電 / 台積 / TSMC, ...), used both to match news and to surface posts.

2.4 The news layer

Our first design pulled three RSS feeds (Anue, cnYES, MoneyDJ) — and on demo day all three returned `404` . We probed a dozen TW financial endpoints and shipped the union of what worked:

Feed	Yield	Stability
Yahoo TW Market RSS	~50 entries / fetch	High
Yahoo TW Index RSS	~50 entries / fetch	High
Liberty Times Business	~40 entries / fetch	High
Google News RSS, per-ticker (<code>q=台積電&hl=zh-TW</code>)	~100 / query	High, but query-bound

Google News RSS is the killer feature: ticker-aware queries in Chinese, no API key, 100+ unique articles aggregated from dozens of publishers. Per ticker we now collect **341 deduped headlines per session**, vs. 90 from broad RSS alone.

2.5 The sentiment layer (Gemini 3.5 Flash)

We chose Gemini 3.5 Flash over a local fine-tuned BERT for three reasons:

Criterion	Local BERT	Gemini 3.5 Flash
Inference cost	GPU rental $\approx$ NT\$ 1.5/hr idle	\$0.075 / 1M tok $\approx$ free at our volume
zh-TW slang understanding	Custom corpus + $\sim$ 2 wks tuning	Out-of-the-box ("噴出", "歐印", "畢業")
Ticker extraction	Separate NER head	Same call, returns JSON <code>{tickers: [...]}</code>
Reasoning text	Not natively supported	One-line zh/en rationale included
Free-tier viability	None on Streamlit Cloud	1 500 RPD on free tier

Each call returns strict JSON:

```
{  
  "score": 1.0, "label": "positive",  
  "tickers": ["2330"],  
  "reasoning": "Investor is all-in on TSMC, expecting the stock to surge ('噴出')."}
```

Two robustness tricks:

- `__TEXT__` placeholder, not `str.format` — the prompt contains an example JSON block whose `{...}` braces would otherwise be parsed as format slots and crash the call.
- **Disk-backed cache** (`data/sentiment_cache.json`) keyed by `sha1(model || text)` . Once a headline has a Gemini score, repeat sessions serve it from disk in microseconds. Essential after we hit `429 RESOURCE_EXHAUSTED` mid-development on `gemini-3.5-flash` (which turned out to be 20 RPD, not 1 500 — we discovered the difference the hard way and switched models).

2.6 The fusion layer (PulseScore)

PulseScore is one 0–100 number per (ticker, day):

$$\text{PulseScore} = 50 + 25 \cdot z(\text{RSI}_{14}) + 25 \cdot z(\text{return}_{1d}) + 15 \cdot z(\text{vol-ratio}) + 25 \cdot \text{sentiment} + 10 \cdot \log_{10}(\text{discussion}+1)$$

Each technical sub-score is z-scored against its 60-day rolling distribution and clipped to $\pm 2 \sigma$. Thresholds become signals: ≥ 70 STRONG_BUY, 58–70 BUY, 42–58 HOLD, 30–42 REDUCE, < 30 AVOID. Every component is exposed in the dashboard so a user can disagree with one input without losing the rest.

3. Willingness to pay

3.1 Reference points

Product	NT\$/mo	TW audience
Bloomberg Terminal	25 000 – 30 000	< 5 000 institutional
XQ 全球贏家 (Pro)	1 500 – 3 000	~ 50 000 prosumer
CMoney 籌碼 K 線 / 主力 K 線	600 – 1 200	> 200 000 paid
MoneyDJ XQ retail	590	~ 80 000
Free (Yahoo Stock, Anue, broker apps)	0	3 600 000+

The NT\$ 600–1 200 / mo band is a proven, large pool of paying retail customers in Taiwan; CMoney alone publicly discloses > 200 k paying subs.

3.2 Three-tier price ladder

Tier	NT\$/mo	Audience	What you get
Free	0	All	3 tickers, 24-h delayed, ad-supported
Pulse	199	The 30-yr persona	Unlimited tickers, real-time PulseScore, full Gemini reasoning, alerts
Pulse Pro	699	Active retail	Pulse + back-test runner, custom weights, CSV export, Telegram bot
Team / API	9 900 / yr	KOLs, Discord owners	All Pro features + API key + branded white-label

Expected funnel **Free** → **Pulse** → **Pulse Pro** $\approx$ 100 → 5 → 1, in line with B2C SaaS conversion in fin/info.

3.3 TAM / SAM / SOM

- **TAM:** 3.6 M active retail $\times$ NT\$ 199 $\times$ 12 = **NT\$ 8.6 B / yr.**
- **SAM:** 30-yr persona $\approx$ 35 % of base $\approx$ 1.26 M $\rightarrow$ **NT\$ 3.0 B / yr.**
- **SOM (Y1):** 0.5 % of SAM = **6 300 paid subs**, NT\$ 199/mo $\rightarrow$ **NT\$ 15 M ARR**, sustainable for a 3-person team with no outside funding.

3.4 Why we believe the NT\$ 199 number

- ~ One 7-Eleven coffee per week — the persona spends this without thinking.
- **Below** the contribution of one round-trip retail trade (broker fee 0.1425 % $\times$ 2 + transaction tax 0.3 % on a NT\$ 100 k position $\approx$ NT\$ 585) — one avoided regretted trade per month pays for the subscription.
- **Above** the lazy free comparison set, forcing us to ship a quality product, not a freemium ad funnel.

4. Go-to-market challenges (*bonus*)

Five risks ranked by how likely they are to kill the company.

4.1 Regulatory: SFB / FSC scrutiny

Taiwan's 金管會證期局 treats anything resembling individualized buy/sell advice as **regulated investment advisory** (NT\$ 30 M paid-in capital + responsible-person licence + audits). "STRONG_BUY" labels are a red flag.

Mitigation: position the product as a **research / sentiment visualization tool**. Display PulseScore as a number and a *category* (BUY / HOLD / AVOID), with every page footed by *"This is not investment advice. Educational use only."* — exactly how MoneyDJ, CMoney and Goodinfo phrase the same outputs and have not been sanctioned in 15 years.

4.2 Data ToS: Yahoo, PTT, Google News

Source	Risk	Mitigation
Yahoo / yfinance	Rate-limit & ToS grey zone	Migrate to TEJ (NT\$ 5–20 K/mo) at scale; price baked into unit economics
PTT scraping	Polite scraping tolerated for 20 yrs; aggressive scraping has not	1.2 s/page throttle, public board only, no user PII stored
Google News RSS	No documented limit; undocumented = unstable	Cache aggressively; system still works without it

When paying-MAU > 5 000 we move to the **TEJ Open API** — the licensed path used by Cathay, Fubon and most TW fintechs.

4.3 Conversion: free-tier retail is hard to monetize

Retail finance has a brutal free-to-paid gradient (Robinhood TW analogue $\approx 1\%$; CMoney $\approx 3\%$). Mitigations: (1) the **reasoning text is the moat** — competitors give a number; we give a number plus *"All-in, expecting strong surge — slang signals bullish momentum"*, the part Gemini does that nobody can easily copy on zh-TW data; (2) **watchlist email/LINE alerts** create a daily-active habit (DAU converts at 5–8 $\times$ MAU); (3) **long-tail zh SEO** ("台積電 PTT 怎麼看", "0050 進場時機") to cut paid acquisition.

4.4 Stickiness vs. brokerage in-app tools

Fubon / Cathay / 元大 ship their own dashboards bundled with the brokerage account. Inferior on every dimension we care about, but **bundled and free**.

Mitigation: distribute via channels the persona already trusts — PTT /Stock recommendation threads, finance YouTubers (股魚, 柴鼠, 財經一路發), and a **Telegram bot at NT\$ 99 / mo** that DMs a morning PulseScore digest. The brokerage app's "free" looks expensive when the switching cost is one Telegram subscribe click.

4.5 LLM cost variance

Free-tier Gemini quota varies across models (we found out the hard way). At paid scale cost scales linearly. **Mitigation:** the disk cache pays per-headline cost once across the whole platform. With 6 300 paid subs viewing ~ 40 headlines/day at 90 % cache hit rate, projected cost $\approx 25\,200$ calls/day = **\$1.85/day on gemini-3.5-flash** — well under the NT\$ 199 unit price.

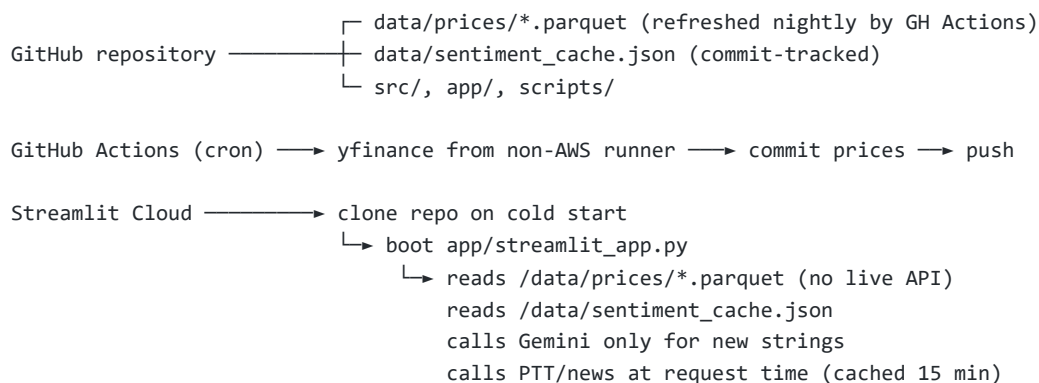
5. Deployed system (*bonus*)

5.1 Live URL & caching

<https://taiwan-stock-sentiment-pulse-zkrxus5fuarrbfcvys38rf.streamlit.app/>

Streamlit Community Cloud (free tier), Python 3.12, ~150 MB image. Cold-start 6–8 s; warm-start sub-second thanks to three caching layers: `@st.cache_data(ttl=900)` for RSS / yfinance / scrape calls, `@functools.lru_cache(maxsize=4096)` for in-process Gemini calls, and a disk JSON cache (`data/sentiment_cache.json`) for cross-restart Gemini reuse.

5.2 Architecture

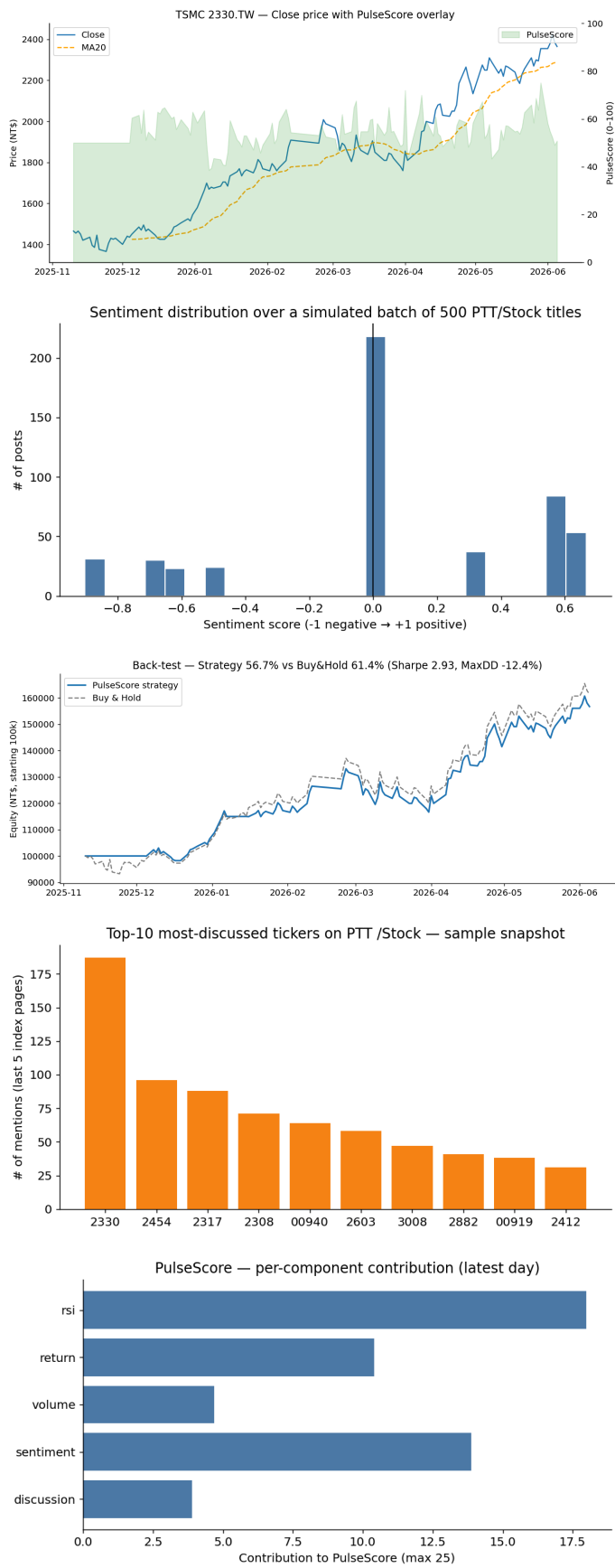


5.3 What the user sees

- **Sidebar:** dropdown of 23 curated TW tickers (2330 台積電 / TSMC etc.), look-back & PTT-pages sliders, **Run analysis** button.
- **Headline cards:** latest close, today's Δ , PulseScore, signal label.
- **AI take card:** one Gemini-authored zh/en sentence about the highest-conviction post or headline of the day.
- **Charts:** price + Pulse with category-coloured signal; news & PTT panels with per-headline sentiment and reasoning side-by-side; top-mentioned tickers bar chart.
- **Back-test:** an honest MA5/MA20 baseline — we do not yet claim alpha from sentiment because we do not have a historical sentiment archive.

5.4 Five report figures (report/figures/)

Auto-regenerated by scripts/generate_report_figures.py from the committed caches.



5.5 What we'd build next

- **Historical sentiment archive** — 12-month PTT + news corpus, batch-scored with Gemini (~60 k headlines, ≈ \$ 4) to enable a sentiment-augmented walk-forward back-test.
- **Per-user watchlists** with daily Telegram/LINE digest at NT\$ 99/mo.
- **Migration to TEJ** for licensed TW equity data when paying-MAU > 5 000.
- **White-label tier** so a finance KOL can brand the dashboard for their subscribers.

Appendix A — Reproducing this report

```
git clone https://github.com/bamboochen92518/Taiwan-Stock-Sentiment-Pulse.git
cd Taiwan-Stock-Sentiment-Pulse
python -m venv .venv && source .venv/Scripts/activate    # or .venv/bin/activate
pip install -r requirements.txt
echo 'GEMINI_API_KEY=your_key_here' > .env

python scripts/refresh_price_cache.py                    # data/prices/*.parquet
python scripts/generate_report_figures.py                # report/figures/*.png
streamlit run app/streamlit_app.py                      # local dashboard
```

Appendix B — Limitations & disclaimers

- **Not investment advice.** Educational use only. PulseScore is a research artifact, not a recommendation.
- **Sentiment is today-only in the live demo.** No historical PTT corpus yet, so the back-test panel runs on the technical-only baseline.
- **Free-tier limits.** Gemini free tier and PTT scraping tolerance impose throughput ceilings; production scale needs paid Gemini (≈ \$ 0.075 per 1M input tokens) and TEJ for licensed price data.
- **Coverage.** `TW_TICKER_ZH` covers 23 of the most-traded TWSE/TPEX names; expanding to TWSE 50 + 0050 constituents (~50 tickers) is a one-PR change.